SAUD AHMAD

AI ENGINEER

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PROFILE

AI Engineer and **co-founder** building AI systems that run in real production, not just prototypes. I am currently an AI Engineer at COTER Global and co-founder of **Oval Labs**, an AI automation studio delivering production AI systems for clients in manufacturing, construction, and recruitment. My work spans the full range of AI engineering, including **LLM integration, retrieval-augmented generation, agent orchestration, prompt and context design, computer vision, and model fine-tuning, backed by full stack engineering and cloud deployment.**

EXPERIENCE

Oval Labs, Co-Founder and AI Engineer **Jan 2026 to Present**

Oval Labs is an AI automation studio that turns manual business workflows into systems that run themselves. The studio designs, builds and maintains production software end to end, from architecture and AI design through to live deployment and ongoing support, across recruitment, aftersales, manufacturing and sales operations.

COTER Global, AI Engineer, Contract (Remote)**Sep 2025 to Present**

- Built the hiring pipeline the firm now recruits through, taking it from first requirement to shortlisted candidate without manual handoffs.
- Own the platform in production: ongoing support, iteration and reliability across live requisitions.

[PRODUCTS OF OVAL LABS]

1. Nebula, AI Recruitment Platform

- Recruitment automation platform: hiring requirements parsed by AI into structured roles, job posts generated automatically across 4 platforms, chatbot-based CV intake, and automated candidate screening and shortlisting.
- Automated WhatsApp and Gmail outreach with AI auto-reply, collecting candidate availability at scale without recruiter follow-up.

2. Centauri, AI Aftersales Platform

- Aftersales support platform covering the full ticket lifecycle: AI issue intake that captures customer issues accurately without data entry staff, a customer portal with real-time status tracking and automated approvals, and AI-assisted dispatch that matches the right technician to each job.
- Field companion that lets technicians log diagnoses by video, voice or text, with required fields populated automatically from the input.
- Natural-language analytics over aftersales data, plus a performance dashboard covering resolution times, team performance and customer satisfaction.

3. ZYP Assembly QC System

- Two-gate quality control system: 14-station in-line inspection with operator checklists and verifier sign-off, followed by a separate pre-delivery inspection gate before a bike ships.
- Live analytics dashboard surfaced a 33% first-pass PDI clearance rate that paper tracking had never made visible, giving the plant its first quantified defect signal.

4. Cold Outreach Platform

- Internal client acquisition system: upload a lead list, AI drafts a personalised cold email per lead, delivered under sending rate limits at up to 150 emails a day.
- Automated 3-step follow-up sequence with a different angle at each step, plus reply and bounce detection sorted into a kanban board.

[FREELANCE WORK, 2025]

TrueSight, AI Deepfake Detection

- 3-layer detection pipeline: a video deepfake model (face detection over 40 sampled frames with a 2-factor consensus rule), an audio deepfake model with segment-by-segment timeline scoring, and a rule-based forensic metadata scanner. None of the three depends on the others.
- Pre-flight quality checks that grade whether an uploaded file is good enough to trust before running full analysis.

Delivered to client.

HSE Performance Tracker (UAE construction firm)

- Multi-tenant safety compliance platform tracking 30 daily HSE fields per worker per project: task briefings, toolbox talks, NCR closures and safety walks.
- Entries roll into a live performance score with monthly KPI tracking and project-level access control, so a site lead only sees their assigned projects. Currently tracking 123 workers across roughly 3,000 man-hours.

Delivered to client.

PROJECTS

Full writeups, code and results for every project are at saudassist.up.railway.app/projects. Section titles below link straight to each collection.

Agentic AI *2 projects*

Two agents that decide for themselves how much work a question deserves. The first reads a question and picks its own depth: answer immediately, run a single bounded search, or launch a full research pass that reads five sources live and writes the findings up as they stream in. The second runs on my portfolio site, choosing turn by turn whether it already knows enough to answer or needs to pull a project's full detail. Both tackle the same problem, which is stopping an agent from spending time and money on questions that never needed it.

RAG and Retrieval *4 projects*

Four studies of one question: how do you get a language model to answer from your own documents and actually be right? I built five retrieval approaches from scratch and ran them against identical questions side by side, which showed that the most sophisticated method was not the best one and documented exactly where two popular techniques broke down. The other three tackle what happens next: trimming retrieved text before it reaches the model so the context budget stretches further, tying every claim in an answer back to the passage it came from so nothing is left unsourced, and testing retrieval against a large public benchmark to see how quality holds up as the document collection grows.

Fine-Tuning and PEFT *1 project*

Adapting a language model to a new task without paying for a full retrain. I implemented the lightweight adaptation method by hand rather than importing a library, then ran it head to head against a complete retrain of the same model on the same task, measuring what you actually give up: training time, storage per saved model, and final quality.

Vision-Language and Multimodal *2 projects*

Getting models to connect images and words. One project searches a real photo library from a plain English description with no tagging, no training and no labels, relying only on the shared meaning between a picture and a sentence. The other is a small image-captioning system built by hand: a vision model and a language model, both left frozen, joined by a small translation layer trained from scratch so that one can speak to the other.

Deep Learning *4 projects*

Four builds spanning security, image generation, text and audio. The security project finds intrusions in network traffic with no labelled attacks to learn from, reaching 0.95 ROC-AUC across 22,544 connections by learning what normal looks like and flagging what isn't. The generation project builds two competing image-generation approaches from scratch and surfaces a real quirk in how one of them reports its own progress. The remaining two classify sentiment across 50,000 film reviews, where a simpler baseline honestly won, and recognise short spoken commands from raw audio.

Machine Learning *6 projects*

Six applied problems, each ending in a decision rather than a leaderboard score. Predicting customer churn across 7,032 accounts, where three models were benchmarked and the simplest won on the evidence at 0.836 ROC-AUC. Catching fraud in 283,000 card transactions, where the textbook approach produced a 95% false alarm rate until the evaluation metric was corrected. Estimating house prices to 0.924 R^2 from 216 engineered features, and forecasting sales demand six months ahead at 83.3% accuracy from four years of history. Two more are honest negative results: heart disease models that came back statistically tied, and a recommendation engine that only narrowly beat a naive baseline. Both are written up that way.

Robotics and IoT *6 projects*

Six systems that had to work in the physical world rather than on a laptop. A solar-powered field robot that identifies and removes weeds on its own, running its vision on-board at 95.6% detection accuracy. A face-recognition entry system deployed at a university campus with 536 real logged events, and a second system that takes attendance from a live video stream. Alongside these, a home automation controller that runs entirely on its own hardware with no cloud service behind it, a rover that samples soil moisture and reports back to a dashboard, and a robot combining obstacle avoidance, line following and remote control on a single chassis.

TECHNICAL SKILLS

AI and LLMs: OpenAI, Gemini, Claude APIs. LangChain, LangGraph. Retrieval-Augmented Generation (RAG) including Naive, Hybrid, Re-ranked, HyDE and Agentic, all built from scratch. FAISS, sentence-transformers, rank-bm25, cross-encoder re-ranking. Agentic tool-use architectures, Server-Sent Events streaming, grounded prompting.

ML and Deep Learning: PyTorch, Hugging Face Transformers, scikit-learn, XGBoost. LoRA built from scratch, GPT-2 fine-tuning. CLIP (ViT-B/32), YOLOv8 and YOLOv8n-face, DeepFace/ArcFace. LSTM, DCGAN, U-Net and diffusion models.

Data and Analytics: pandas, NumPy, SciPy with statistical significance testing, SHAP, statsmodels (SARIMA), Prophet, matplotlib. Feature engineering and imbalanced-data handling with SMOTE and alternatives.

Backend and Frontend: Python, C/C++, FastAPI, Next.js, React, WebSocket. PostgreSQL, SQLite, SQLAlchemy, Redis. JWT auth and role-based access control, Tailwind CSS.

Infrastructure and Deployment: Docker, Railway, Cloudflare R2. Google Sheets API, Gmail SMTP/IMAP, WhatsApp (Baileys), LinkedIn and Meta Graph API.

Robotics and Embedded: Jetson Orin Nano, Arduino (Mega, WiFiS3), ESP32, Raspberry Pi. GPS and IMU sensor fusion, OpenCV, face_recognition (dlib), MQTT, Blynk IoT, Bluetooth Serial.

EDUCATION

BSc Data Science, GIK Institute of Engineering Sciences and Technology2021 to 2025

Coursework: Machine Learning, Deep Learning, NLP, Data Engineering, Database Systems, Software Engineering.

Technical Lead, Team Techno (Robotics)

Led GIK Institute's robotics team and set technical direction across hardware and software. That team is the foundation behind AgroBot, Access Vision and the other robotics builds here.

CERTIFICATIONS

IBM: Machine Learning with Python | IBM: Python for Data Science and AI | IBM: Data Science Methodology | DeepLearning.AI: Generative AI for Everyone